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HISTORY · Mughal Empire: administration, society, economy

UNDERSTAND IT FIRST

What is it? The **Mughals** (1526–1857; real power until **1707**) built the largest and richest empire in medieval India. Their great gift was an **administrative system** (mansabdari, provinces, land revenue) that later rulers, including the **British**, copied. They also joined **Persian and Indian** styles in art and architecture and, under **Akbar**, tried a policy of **religious tolerance**. Learn it as **Sources → Rulers → Administration → Politics · Society · Economy · Religion → Art and architecture → Decline**.

1. Sources

Source	Author and what it tells
Baburnama (Tuzuk-i-Baburi)	Babur's memoir in **Turki**; India, its people and wildlife.
Humayunnama	**Gulbadan Begum**, Humayun's sister.
Akbarnama, **Ain-i-Akbari**	**Abul Fazl**; Akbar's reign; the Ain has statistics on administration, revenue, army, provinces.
Tuzuk-i-Jahangiri	Jahangir's memoir (Persian).
Padshahnama	**Abdul Hamid Lahori**; Shah Jahan.
Other Persian works	**Muntakhab-ut-Tawarikh** (Badauni, critic of Akbar), **Tabaqat-i-Akbari** (Nizamuddin Ahmad).
Foreign travellers	**Hawkins, Thomas Roe** (English, Jahangir); **Tavernier, Bernier** (French); **Manucci** (Italian); **Peter Mundy**; Jesuit **Monserrate** (Akbar).

2. The rulers, in order

Ruler	Years (CE)	Key facts
Babur	1526–30	Defeated Ibrahim Lodi at the **First Battle of Panipat (1526)** using **artillery** and the **tulughma** tactic; **Khanwa (1527)** vs Rana Sanga; **Chanderi (1528)**; **Ghagra (1529)** vs Afghans.
Humayun	1530–40, 1555–56	Lost to **Sher Shah** at **Chausa (1539)** and **Kannauj / Bilgram (1540)**; exile in Persia; regained Delhi in **1555**; died after a fall in his library.

Sher Shah Suri	1540–45	The **Sur** interlude. **Reforms:** land **measurement**, **patta and qabuliyat**, silver **rupiya** (178 grains) and copper **dam**, **Grand Trunk Road** with **sarais** and **dak chowkis**, **sarkar–pargana** divisions, the **Rohtas fort** and **Purana Qila**; died at **Kalinjar (1545)**. Akbar built on his system.
Akbar	1556–1605	Won the **Second Battle of Panipat (1556)** against Hemu under regent **Bairam Khan** (to 1560). Conquests: **Malwa**, **Gondwana** (Rani **Durgavati**, 1564), **Chittor (1568)**, **Ranthambore (1569)**, **Gujarat (1572–73)**, **Bengal (1574–76)**, **Haldighati (1576)** vs Rana Pratap, **Kashmir (1586)**, Sindh, Orissa, **Khandesh and Asirgarh (1601)**; **Fatehpur Sikri** (capital from 1571); **Rajput alliance** (marriages, high posts); **Mansabdari**, **Dahsala**; **Sulh-i-kul**; **Din-i-Ilahi (1582)**.
Jahangir	1605–27	Rebellion of son **Khusrau**; execution of Sikh Guru **Arjan** (1606); the **Chain of Justice**; **Captain Hawkins (1608)** and **Sir Thomas Roe (1615)**; treaty with **Mewar (1615)**; **Nur Jahan** and her **junta** held great power; painting at its peak (Ustad **Mansur**, **Abul Hasan**).
Shah Jahan	1628–58	Architecture at its height (**Taj Mahal**, **Red Fort**, **Jama Masjid** Delhi, **Peacock Throne**); annexed **Ahmednagar (1633)**; treaties with **Golconda and Bijapur (1636)**; the war of succession among his sons (**1657–58**) won by **Aurangzeb**; confined at Agra.
Aurangzeb	1658–1707	Called **Alamgir**; **reimposed jizya (1679)**; executed Sikh Guru **Tegh Bahadur (1675)**; revolts of **Jats, Satnamis, Rajputs, Sikhs** and **Marathas** (**Shivaji**); conquered **Bijapur (1686)** and **Golconda (1687)**, taking the empire to its widest extent, but 25 years in the **Deccan** drained the treasury; died at Ahmednagar (1707). Built the **Moti Masjid**, **Badshahi Mosque** (Lahore).
Later Mughals	1707–1857	Bahadur Shah I; **Farrukhsiyar** (1717 **farman** gave the English East India Company trade privileges in Bengal); **Nadir Shah** sacked Delhi (**1739**, took the Peacock Throne and Kohinoor); **Ahmad Shah Abdali**, Third Panipat (**1761**); **Shah Alam II** defeated at **Buxar (1764)**; last emperor **Bahadur Shah II Zafar** exiled after **1857**.

EXAMPLE

Deccan and Andhra link (all regions equal): Shah Jahan's treaty (1636) made **Golconda** (Qutb Shahi, Hyderabad) a tributary; **Mir Jumla**, a Golconda noble who helped conquer the Karnatak coast, joined **Aurangzeb**; Aurangzeb took **Golconda in 1687** after a long siege; **Masulipatnam** was the chief east-coast port for cloth; the **Nizam-ul-Mulk Asaf Jah** (1724) founded the Hyderabad state as Mughal power faded.

3. Administration

Mughal administration

Level	Details

Emperor	Absolute head; **Padshah**, **Zill-i-Ilahi** (shadow of God). Akbar claimed **divine light (farr-i-izadi)**; daily **jharokha darshan**, **durbar**; Akbar's Persian ideas of kingship by Abul Fazl.
Central ministers	**Wakil** (prime minister; later the **Wazir / Diwan-i-Kul** for revenue and finance); **Mir Bakshi** (military pay and mansabs); **Mir Saman** (royal household and **karkhanas**); **Sadr-us-Sudur** (charity and religious grants: **madad-i-maash / suyurghal**); **Qazi-ul-Quzat** (chief judge); **Muhtasib** (public morals, under Aurangzeb); **Daroga-i-Dak Chowki**.
Subah (province)	**12 subahs** (Ain, 1580), later **15** under Akbar, **21** under Aurangzeb. **Subadar** (governor), **Diwan** (revenue; named by the emperor, independent of the Subadar), **Bakshi**, **Sadr**, **Qazi**, **Waqia-navis** (reporter): a **system of checks**.
Sarkar, Pargana, Village	**Sarkar**: **Faujdar** (law and order, army) and **Amalguzar** (revenue). **Pargana**: **Shiqdar**, **Amin**, **Qanungo** (land records), **Karkun**. **Village**: **Muqaddam** (headman), **Patwari** (accountant), **zamindars**.

The Mansabdari system

- **Mansab** = rank. Introduced by **Akbar** (c. **1570–71**); every state officer, civil or military, held a mansab. Rank by **zat** (personal rank, which fixed **pay** and status) and **sawar** (the number of **cavalrymen** he had to keep). Ranks from **10 to 5,000** for nobles (later to 7,000 and 10,000; princes up to 12,000). Three groups: **mansabdar** (up to 500 zat), **amir** (500–2,500), **amir-i-umda** (above 2,500).
- **Not hereditary**; the emperor appointed, promoted, demoted and transferred them. **Pay**: in cash (**naqdi**) or by a **jagir** (assignment of the **revenue** of a region, not ownership; changed every 2–3 years so nobles could not build local power). **Dagh** (branding horses) and **chehra** (descriptive roll) checked fraud. **Sawar** ranks: if 'sawar' was equal to zat = first class; half or less = second/third class.
- **Later problem:** by **Aurangzeb** there were too many mansabdars and too few jagirs (**jagirdari crisis**, **bejagiri**), so nobles squeezed the peasants and the army weakened.

Land revenue

System	How it worked
Zabt / Dahsala (Todar Mal; 1580)	Land was **measured** (with a **jarib**, bamboo joined by iron rings) and classed as **Polaj** (cultivated every year), **Parauti** (fallow 1–2 years), **Chachar** (fallow 3–4) and **Banjar** (uncultivated 5+). The **average yield and prices of the last ten years** were used to fix the **state's share: one-third** of the average produce, paid in **cash**, in a fixed rate per **bigha** for each crop in each area (**dastur**). Used mainly in **Delhi, Agra, Lahore, Multan, Malwa**.
Batai / Ghallabakshi	Crop-sharing at harvest; **Kankut** (estimate of the standing crop); **Nasaq** (an estimate from past records).
Other taxes	**Jizya** (abolished by Akbar in **1564**, reimposed by Aurangzeb in **1679**); pilgrim tax abolished **1563**; **zakat** on Muslim traders; customs and transit duties (**rahdari**). In practice the state share was about **one-third to one-half**.

4. Politics · Society · Economy · Religion of the Mughals

Area	Mughal period
Politics	**Centralised absolute monarchy** with a paid, rotating nobility (**mansabdars**, not hereditary); **Rajput alliance** (Akbar married Amber's princess; Rajputs like **Man Singh** and **Todar Mal** held top posts) widened the base; **Sulh-i-kul**; a **war of succession** at nearly every change; the **Deccan** question; **decline** after Aurangzeb from **jagirdari crisis**, revolts, weak successors and invasions (**Nadir Shah 1739**).
Society	**Nobility**: Turani, Irani, Afghan, Rajput, later Deccani and Maratha; **zamindars** (hereditary local lords with forts and troops) shared the revenue; **peasants**: **khud-kasht** (resident) and **pahi-kasht** (non-resident); **artisans** and a growing **merchant** class; **women**: royal women like **Nur Jahan**, **Jahanara** had influence, but **purdah** and **sati** persisted (Akbar discouraged compulsory sati and child marriage and allowed widow remarriage); **slavery** continued; **peasant revolts**: **Jat** (Gokula, 1669), **Satnami** (1672), **Sikh**; **famines** (1595–98, 1630–32).
Economy	**Agriculture**: wheat, rice, millets, **cotton**, **indigo** (Bayana), sugarcane, opium; **new crops** came with the Portuguese: **tobacco**, **maize**, **potato**, **chilli**, **pineapple**. **Crafts**: **cotton textiles** (Dhaka muslin, **Masulipatnam** calico, Burhanpur), silk, shawls (Kashmir), carpets, ships, arms; royal **karkhanas**. **Currency**: silver **rupiya**, copper **dam**, gold **mohur**; a **cash economy**; **hundis** (bills of exchange) and **sarrafs** (bankers). **Trade**: inland by **banjaras**; ports **Surat** (gateway to Mecca), **Masulipatnam**, **Hooghly**, Cambay; **European companies** (Portuguese, **English 1600**, **Dutch 1602**, **French 1664**) brought **silver**, so the balance of trade favoured India. **Cities**: Agra, Delhi, Lahore, Fatehpur Sikri, Dhaka, Ahmedabad.
Religion	**Akbar**: abolished the **pilgrim tax (1563)** and **jizya (1564)**; **Ibadat Khana (1575)** at Fatehpur Sikri for debates with Ulema, Hindus, Jains, Parsis and **Jesuits**; the **Mahzar (1579)** made him arbiter of religious disputes; **Din-i-Ilahi (1582)**, a small ethical order, not a new religion; policy of **Sulh-i-kul** (peace with all). **Jahangir** continued tolerance; **Shah Jahan** was more orthodox; **Aurangzeb** reimposed jizya (1679), banned court music and tried to enforce the sharia. **Sufis**: Chishti (Salim Chishti), **Naqshbandi** (**Shaikh Ahmad Sirhindi**, who opposed Akbar's ideas). **Bhakti**: **Tulsidas** (*Ramcharitmanas*), **Surdas**, **Mirabai**, **Tukaram**, **Eknath**, **Namdev**. **Sikhism**: Guru **Nanak**, Guru **Arjan** built the **Harmandir Sahib** and compiled the **Adi Granth**, Guru **Hargobind** raised an army, Guru **Gobind Singh** founded the **Khalsa (1699)**.
Art, architecture and literature	**Architecture** (Persian + Indian): **Humayun's Tomb** (first **garden tomb**, double dome, red sandstone and marble), **Agra Fort**, **Fatehpur Sikri** (Buland Darwaza, Jama Masjid, Panch Mahal, Diwan-i-Khas, Tomb of Salim Chishti), **Akbar's tomb at Sikandra**, **Itimad-ud-Daula** (first use of **pietra dura**), **Taj Mahal** (1632–53, Ustad **Ahmad Lahori**, white marble, **charbagh**, pietra dura), **Red Fort** (1648), **Jama Masjid**. **Painting**: Persian + Indian miniatures: **Hamzanama**, **Akbarnama** illustrations; **Jahangir's** nature and portrait paintings; Ustad **Mansur**, **Abul Hasan**. **Music**: **Tansen**, **Dhrupad**. **Literature**: Persian; Hindi (Tulsidas, Surdas, Keshavdas); **Urdu** developed.

5. Decline

Causes: the **jagirdari crisis**, **Aurangzeb's Deccan wars** and religious policy (jizya, conflict with Rajputs, Sikhs and Marathas), **peasant revolts**, weak successors and noble factions, **Nadir Shah (1739)** and **Abdali (1761)**,

the rise of **regional states** (Awadh, Bengal, Hyderabad, Marathas) and finally the **East India Company** (Plassey 1757, Buxar 1764).

EXAM TRAP

Panipat I 1526 (Babur), **II 1556** (Akbar), **III 1761** (Abdali vs Marathas); **Akbar**: mansabdari c. 1570, Dahsala 1580, jizya abolished 1564, Ibadat Khana 1575, Din-i-Ilahi 1582, Fatehpur Sikri, Buland Darwaza 1573; **Jahangir**: Chain of Justice, Roe 1615; **Shah Jahan**: Taj Mahal, Golconda treaty 1636; **Aurangzeb**: jizya 1679, Golconda 1687; **zat = rank and pay, sawar = cavalry**; mansabs **not hereditary**; **jagir = revenue, not land ownership**; **Zabt = one-third in cash**.

Test yourself: 5 questions (tap an option to see the answer)

PRACTICE

Consider the following statements about the Mughal mansabdari system:
1. It was introduced by Akbar.
2. Mansabs were hereditary.
3. 'Zat' denoted the personal rank and pay, 'sawar' the number of cavalrymen to be maintained.
4. A jagir was an assignment of land revenue, not ownership of the land.
Which of the above are correct?

A. 1, 3 and 4 only

B. 1 and 2 only

C. 2, 3 and 4 only

D. 1, 2, 3 and 4

Answer: A. Mansabs were not hereditary; the emperor controlled them.

PRACTICE

Match List-I (work) with List-II (author/subject):
List-I: A. Ain-i-Akbari B. Humayunnama C. Padshahnama D. Baburnama
List-II: 1. Abdul Hamid Lahori 2. Gulbadan Begum 3. Abul Fazl 4. Babur

A. A-3, B-2, C-1, D-4

B. A-2, B-3, C-1, D-4

C. A-3, B-2, C-4, D-1

D. A-1, B-2, C-3, D-4

Answer: A. Ain: Abul Fazl; Humayunnama: Gulbadan; Padshahnama: Lahori; Baburnama: Babur.

PRACTICE

Which one of the following statements is NOT correct?

A. Todar Mal's Dahsala system used the average produce and prices of ten years.

B. Under the zabt system land was classed as polaj, parauti, chachar and banjar.

C. Akbar abolished jizya in 1564, and it was never imposed again.

D. The state's share under zabt was about one-third of the average produce, paid in cash.

Answer: C. Aurangzeb reimposed jizya in 1679.

PRACTICE

Assertion (A): The mansabdari system helped the Mughal emperor keep control over the nobility.
Reason (R): Mansabs were not hereditary and the emperor could promote, demote and transfer mansabdars and their jagirs.
Choose the correct answer:

A. Both A and R are true and R is the correct explanation of A

B. Both A and R are true but R is not the correct explanation of A

C. A is true but R is false

D. A is false but R is true

Answer: A. Dependence on the emperor for rank and jagir kept the nobles loyal.

PRACTICE

A gateway about 54 metres high at Fatehpur Sikri commemorates the conquest of Gujarat. Which emperor built it?

A. Akbar

B. Shah Jahan

C. Jahangir

D. Humayun

Answer: A. The Buland Darwaza was built by Akbar c. 1573.

RECALL BOX: CLOSE THE NOTES AND WRITE THESE IN YOUR OWN WORDS

1. Write the 8 Mughal rulers with key facts. 2. Draw the administration chart and explain mansabdari, zabt and jagir. 3. Write the Politics–Society–Economy–Religion table.

POLITY · Legislature, Executive and Judiciary; unicameral and bicameral legislatures

UNDERSTAND IT FIRST

What is it? A government has three jobs: **make** the laws (**legislature**), **carry them out** (**executive**), and **judge** disputes and test whether laws follow the Constitution (**judiciary**). The French thinker **Montesquieu** said these should be in **different hands** so that no one gets too much power. India follows this in **spirit** but not in a strict way.

Three organs of government

1. Separation of powers in India

- **Not a strict separation** (unlike the USA). The Supreme Court said in **Ram Jawaya Kapur (1955)** that India does not have a rigid separation, but it has a **separation of functions**. The **overlaps:** ministers (executive) are **members of Parliament** and answer to it; the **President** is part of Parliament and the head of the executive; the legislature can **impeach** judges and amend the Constitution; the judiciary makes law through interpretation.
- **Checks and balances: judicial review** (Arts 13, 32, 226); **Parliament controls the executive** by questions, motions, committees and **no-confidence**; the **executive** can issue **ordinances** (Art 123) and appoints judges; **Art 50** directs the State to separate the judiciary from the executive in public services; **Art 121** bars discussion of a judge's conduct in Parliament. Separation of powers is held to be part of the **basic structure**.

2. The Legislature: unicameral and bicameral

A legislature with **one house** is **unicameral**; with **two houses** it is **bicameral**. **Why two?** A second house gives **second thought**, represents **States** or experts and prevents hasty laws. **Why one?** It is simpler, cheaper and avoids delay. **Parliament is bicameral** (**Lok Sabha** = lower house of the people; **Rajya Sabha** = upper house of the States). Parliament = **President + Lok Sabha + Rajya Sabha** (Art 79).

Point	Lok Sabha	Rajya Sabha
Strength	Maximum **552** (530 States + 20 UTs + 2 Anglo-Indian earlier); now **543** elected	Maximum **250**: **238** elected by State/UT legislators + **12 nominated** (art, science, literature, social service); at present 245

Election	**Direct** by the people (adult suffrage, 18+)	**Indirect** (single transferable vote, proportional representation)
Term	**5 years** (can be dissolved earlier)	**Permanent**; members serve **6 years**, **one-third retire every 2 years**
Presiding officer	**Speaker** (elected by members)	**Vice-President** is the ex-officio **Chairman**; a Deputy Chairman is elected
Money Bills	Introduced **only** here; **decides**	Can only **recommend** changes within **14 days**
Council of Ministers	**Responsible** to it; **no-confidence motion** only here	Not responsible; can discuss and ask questions
Special powers	Passes the **budget**; Money Bills	**Art 249**: can authorise Parliament to make a law on a **State List** subject (2/3 present and voting); **Art 312**: create **All India Services**
Joint sitting	**Art 108**: for **ordinary or financial** Bills when the Houses disagree; presided by the **Speaker**; the majority of the larger House (Lok Sabha) usually prevails. **Not** for Money Bills or **Constitution amendment Bills**.	—

State legislatures

- Every State has a **Legislative Assembly** (**Vidhan Sabha**, directly elected, 60 to 500 members, 5 years). **Art 168** allows a second house, the **Legislative Council (Vidhan Parishad)**: only **six States have one**: **Andhra Pradesh, Telangana, Karnataka, Maharashtra, Bihar and Uttar Pradesh**. (J&K's was abolished in 2019.)
- **Creation or abolition (Art 169):** the **Assembly passes a resolution** by a **special majority** (majority of total members and 2/3 present and voting), then **Parliament passes a law by simple majority**. It is **not an amendment** under Art 368. (The Andhra Pradesh Assembly's 2020 resolution to abolish its Council was later withdrawn.)
- **Size of Council:** not more than **one-third** of the Assembly and not less than **40**. **Composition: 1/3** elected by MLAs; **1/3** by local bodies (municipalities, panchayats); **1/12** by graduates; **1/12** by teachers; **1/6** nominated by the **Governor** (art, science, literature, social service). Council is **permanent**, **one-third retire every 2 years**, term **6 years**; it can only **delay** an ordinary Bill (up to **3 months**, then **1 month** second time); Money Bills: only the Assembly decides.

3. The Executive

Level	Details
Union	**President** (nominal head, Arts 52–53; elected indirectly for 5 years by an **electoral college**); **Vice-President**; **Prime Minister and Council of Ministers** (Arts 74–75) **aid and advise** the

	President and are **collectively responsible** to the **Lok Sabha**. The 91st Amendment (2003) limits the Council to **15%** of the Lok Sabha's strength. **Attorney General** (Art 76).
State	**Governor** (appointed by the President; Arts 153–154); **Chief Minister and Council of Ministers** (Art 163–164), collectively responsible to the **Assembly**; **Advocate General**.
Permanent executive	The **civil services** (bureaucracy) carry out policy and continue when governments change.

4. The Judiciary

- India has a **single integrated judiciary**: **Supreme Court (Art 124; 34 judges including the CJI)** at the top, then **25 High Courts** (Art 214), then **district and subordinate courts** (Arts 233–237). (The USA has separate federal and State courts.)
- **Independence:** judges are appointed by the President after consulting the **collegium** (Second Judges case 1993); **security of tenure** (removal only by **impeachment**, Art 124(4), by special majority in both Houses); **salaries charged on the Consolidated Fund**; no discussion of conduct in Parliament (Art 121); power to punish for **contempt**; no practice after retirement in the same court.
- **Functions:** original, appellate and advisory (Art 143) jurisdiction; **judicial review**; guardian of the Constitution and Fundamental Rights (Arts 32, 226); **Lok Adalats** and **tribunals** give quick justice.

EXAM TRAP

Parliament = President + LS + RS (Art 79); **Art 108 joint sitting** (not for Money or Constitution amendment Bills); **Money Bill only in LS**; **no-confidence only in LS**; **RS: permanent, 1/3 retire every 2 years, Art 249**; **Art 168/169**: Councils in **6 States** (AP, Telangana, Karnataka, Maharashtra, Bihar, UP); Council size ≤ 1/3 of Assembly, ≥ 40; **Art 169** by Parliament's simple-majority law, not Art 368; **91st Amendment: ministers ≤ 15%**; **Ram Jawaya Kapur**: no rigid separation.

Test yourself: 5 questions (tap an option to see the answer)

PRACTICE

Consider the following statements about State Legislative Councils:
1. Article 169 provides for the creation or abolition of a Legislative Council.
2. At present Legislative Councils exist in six States.
3. The number of members of a Council cannot exceed one-third of the Assembly and cannot be less than 40.
4. All members of the Council are directly elected by the people.
Which of the above are correct?

A. 1, 2 and 3 only

B. 1 and 4 only

C. 2, 3 and 4 only

D. 1, 2, 3 and 4

Answer: A. Council members are elected indirectly or nominated, not directly by the people.

PRACTICE

Match List-I (Article) with List-II (subject):
List-I: A. Article 108 B. Article 169 C. Article 124 D. Article 74
List-II: 1. Joint sitting of Parliament 2. Creation or abolition of State Legislative Councils 3. Establishment of the Supreme Court 4. Council of Ministers to aid and advise the President

A. A-1, B-2, C-3, D-4

B. A-2, B-1, C-3, D-4

C. A-1, B-2, C-4, D-3

D. A-3, B-2, C-1, D-4

Answer: A. 108: joint sitting; 169: Councils; 124: Supreme Court; 74: Council of Ministers.

PRACTICE

Which one of the following statements is NOT correct?

A. A Money Bill can be introduced only in the Lok Sabha.

B. The Rajya Sabha is a permanent House not subject to dissolution.

C. A no-confidence motion against the Council of Ministers can be moved only in the Lok Sabha.

D. The Rajya Sabha has equal powers with the Lok Sabha on Money Bills.

Answer: D. In Money Bills the Rajya Sabha can only recommend changes within 14 days.

PRACTICE

Assertion (A): India does not have a strict separation of powers.
Reason (R): The ministers who form the executive are also members of the legislature and are responsible to it.
Choose the correct answer:

A. Both A and R are true and R is the correct explanation of A

B. Both A and R are true but R is not the correct explanation of A

C. A is true but R is false

D. A is false but R is true

Answer: A. The parliamentary system itself joins the executive and the legislature.

PRACTICE

The Legislative Assembly of a State passes, by a special majority, a resolution to abolish its Legislative Council. What more is needed to abolish it?

A. A law passed by Parliament by simple majority

B. A Constitution amendment ratified by half of the States

C. The consent of the Governor alone

D. A Supreme Court order

Answer: A. Article 169: Parliament makes the law by simple majority; it is not an Art 368 amendment.

RECALL BOX: CLOSE THE NOTES AND WRITE THESE IN YOUR OWN WORDS

1. Draw the three organs and write the overlaps and checks. 2. Write the Lok Sabha vs Rajya Sabha table. 3. Write Art 168–169 and the Council's composition; list the six States.

MATHS · Number system and order of magnitude

UNDERSTAND IT FIRST

What is it? Everything in arithmetic rests on **numbers**, so first learn the **families** of numbers and their **properties** (even, odd, prime, factors), then the **rules** that speed up calculations (divisibility), and finally how to judge the **size** of a number (order of magnitude), which helps you **estimate** answers quickly in the exam.

Families of numbers

1. Kinds of numbers

Kind	Meaning	Examples / facts
Natural (N)	Counting numbers	1, 2, 3 …

Whole (W)	Natural numbers and 0	0, 1, 2 …
Integers (Z)	Whole numbers and their negatives	… −2, −1, 0, 1 …
Rational (Q)	Written as **p/q** (q ≠ 0); decimals **end** or **repeat**	1/2 = 0.5; 1/3 = 0.333…; 2.75
Irrational	**Cannot** be written as p/q; decimals neither end nor repeat	√2, √3, π, e; the sum of a rational and an irrational is irrational
Real (R)	Rational and irrational together	Every point of the number line
Even / odd	Divisible by 2 or not	even + even = even; odd + odd = even; odd × odd = odd
Prime	Exactly **two** factors: 1 and itself; **1 is neither prime nor composite**; **2 is the only even prime**	**25 primes below 100**: 2, 3, 5, 7, 11, 13, 17, 19, 23, 29, 31, 37, 41, 43, 47, 53, 59, 61, 67, 71, 73, 79, 83, 89, 97
Composite, **co-prime**	Composite: more than 2 factors. Co-prime: **HCF = 1** (need not be prime themselves)	(8, 15) are co-prime; twin primes: (11, 13), (17, 19)
Perfect number	Equal to the sum of its proper divisors	6 = 1+2+3; 28

2. Divisibility rules

By	Rule	Example
2, 5, 10	Last digit even; 0 or 5; 0	3,450 by 2, 5, 10
3, 9	**Sum of digits** divisible by 3 or 9	6,174: 6+1+7+4 = 18, so by 9
4, 8	Last **2** digits by 4; last **3** digits by 8	5,216: 16 by 4 ✓; 216 by 8 ✓
6, 12, 15	6 = 2 and 3; 12 = 3 and 4; 15 = 3 and 5	—
7	Double the last digit, **subtract** from the rest; repeat	203: 20 − 6 = 14 ✓
11	(Sum of digits in odd places) − (sum in even places) is **0 or a multiple of 11**	2,728: (2+2) − (7+8) = −11 ✓
13	Add **4 × last digit** to the rest; repeat	273: 27 + 12 = 39 ✓

3. Factors and divisors

EXAMPLE

Write N as a product of primes: **N = a^p × b^q × c^r**. Then: **number of divisors = (p+1)(q+1)(r+1)**; **sum of divisors = (a⁰+...+a^p)(b⁰+...+b^q)(c⁰+...+c^r)**. A number has an **odd number of divisors only if it is a perfect square**. A number with **exactly 3 divisors** is the **square of a prime**. **Example: 360 = $2^3 \times 3^2 \times 5$**: divisors = 4 × 3 × 2 = **24**; sum = 15 × 13 × 6 = **1,170**; **odd divisors** come from $3^2 \times 5$ only: 3 × 2 = **6**, so **even divisors = 24 − 6 = 18**. **HCF × LCM = product of the two numbers.**

4. Place value and recurring decimals

Indian system	International system	Power of 10
1 lakh = 1,00,000	100 thousand	10^5
10 lakh	1 million = 10,00,000	10^6
1 crore = 1,00,00,000 (100 lakh)	10 million	10^7
1 arab = 100 crore	1 billion	10^9
1 lakh crore	1 trillion	10^{12}

Recurring decimals to fractions: $0.\overline{x}$ (one repeating digit) = x/9; **$0.\overline{xy}$** = xy/99. So 0.333... = 3/9 = 1/3 and 0.4545... = 45/99 = 5/11. For a **mixed** recurring decimal like 0.1666...: let x = 0.1666..., then 10x = 1.666..., 100x = 16.666..., so 90x = 15, x = 1/6.

5. Order of magnitude

UNDERSTAND IT FIRST

Meaning: the **order of magnitude** tells **how big** a number is **in powers of 10**. Write the number as **a × 10ⁿ** with **1 ≤ a < 10**; **n** is its order of magnitude (a stricter physics rule rounds up to n + 1 when a is more than about 3.16, but for the exam use the simple rule). **Shortcut:** order = **(number of digits) − 1** for numbers 1 or more. It lets you **compare** huge numbers and **check** whether an answer is sensible.

Quantity (approximate)	Scientific form	Order
4,86,000 (4.86 lakh)	4.86×10^5	5
India's population ≈ 140 crore	1.4×10^9	9
A State with ≈ 5 crore people	5×10^7	7
Seconds in a day: 86,400	8.64×10^4	4
Earth–Sun distance ≈ 15 crore km	1.5×10^8 km	8

EXAMPLE

Using it: a **ratio** of two numbers has order = difference of the orders. India ≈ 10^9 and a State ≈ 10^7: ratio ≈ 10^2, and exactly $1.4 \times 10^9 \div 5 \times 10^7$ = **28**, between 10^1 and 10^2, about 10^1. **Estimating:** 49 × 51 ≈ 50 × 50

= 2,500 (order 3); 3,98,000 ÷ 19,900 ≈ 4,00,000 ÷ 20,000 = **20**. When options are far apart, **estimation alone** finds the answer.

EXAM TRAP

1 is neither prime nor composite; **2 is the only even prime**; **25 primes below 100**; divisors count = **(p+1)(q+1)...**; **3 divisors ⇒ square of a prime**; **HCF × LCM = product**; **$0.\overline{x} = x/9$**; **1 crore = 10 million = 10^7**; **1 lakh crore = 1 trillion**; order of magnitude = **(digits − 1)**.

Test yourself: 5 questions (tap an option to see the answer)

PRACTICE

How many natural numbers less than 100 have exactly three divisors?

A. 3

B. 4

C. 5

D. 6

Answer: B. Only squares of primes have 3 divisors: 2^2, 3^2, 5^2, 7^2 = 4, 9, 25, 49.

PRACTICE

How many of the divisors of 360 are even?

A. 12

B. 15

C. 18

D. 24

Answer: C. 360 = $2^3 \times 3^2 \times 5$ has 24 divisors; the odd ones come from $3^2 \times 5$ = 6 divisors; even = 24 − 6 = 18.

PRACTICE

What is the smallest six-digit number divisible by 7?

A. 1,00,001

B. 1,00,002

C. 1,00,003

D. 1,00,007

Answer: B. 7 × 14,285 = 99,995; the next multiple 99,995 + 7 = 1,00,002.

PRACTICE

India's population is about 1.4×10^9 and the population of a State is about 5×10^7. The ratio of the two is nearest to which order of magnitude?

A. 10^0

B. 10^1

C. 10^3

D. 10^5

Answer: B. $1.4 \times 10^9 \div 5 \times 10^7 = 28$, which is of the order 10^1.

PRACTICE

What is the value of 0.4545... + 0.2727... (both decimals repeating) as a fraction?

A. 8/11

B. 72/100

C. 7/11

D. 9/11

Answer: A. 45/99 + 27/99 = 72/99 = 8/11.

RECALL BOX: CLOSE THE NOTES AND WRITE THESE IN YOUR OWN WORDS

1. Draw the nested sets and write the prime facts. 2. Write all divisibility rules from 2 to 13 and the divisor formulas. 3. Write the Indian–international table and the order-of-magnitude rule with 2 examples.

NOTEBOOK PAGE · Day 9: write this from memory

Close the notes. Write these headings, fill every blank, then correct in a different colour.

HISTORY: The Mughals, Politics · Society · Economy · Religion

Area	Write
Sources	Baburnama, Humayunnama (Gulbadan), Akbarnama and Ain-i-Akbari (Abul Fazl), Tuzuk-i-Jahangiri, Padshahnama (Lahori), travellers: Roe, Tavernier, Bernier
Rulers	Babur 1526 → Humayun → (Sher Shah) → Akbar 1556 → Jahangir 1605 → Shah Jahan 1628 → Aurangzeb 1658–1707; later Mughals
Politics	Centre: Wakil/Diwan, Mir Bakshi, Mir Saman, Sadr, Qazi; Subah → Sarkar → Pargana → Village; mansabdari (zat, sawar); jagir; dagh
Society	Nobles (Turani, Irani, Afghan, Rajput, Deccani); zamindars; khud-kasht and pahi-kasht peasants; women; revolts: Jat, Satnami, Sikh
Economy	Zabt/Dahsala (Todar Mal), 1/3 revenue in cash; rupiya, dam, mohur; crops; textiles; karkhanas; Surat, Masulipatnam; hundis; European companies
Religion and art	Sulh-i-kul, Ibadat Khana 1575, Mahzar 1579, Din-i-Ilahi 1582; jizya abolished 1564, reimposed 1679; Sikh Gurus; Humayun's Tomb, Fatehpur Sikri, Taj Mahal

POLITY: Three organs; unicameral and bicameral

- Separation of powers in India: functional not strict; examples of overlap and checks; Parliament = President + LS + RS; LS vs RS table; Art 169 Councils (6 States).
- Executive: President, PM and Council of Ministers (91st Amendment 15%); Judiciary: SC, HC, subordinate courts; independence features.

MATHS: Number system and order of magnitude

Draw the nested sets $N \subset W \subset Z \subset Q \subset R$; write divisibility rules for 2 to 13, the number-of-divisors formula, primes below 100 (25), the Indian and international place systems and the order-of-magnitude rule with one example.